

2021
Higher School Certificate
Year 11 Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student number and/or name at the top of every page

Total marks – 100

Section I – Pages 2–9

20 marks

- Attempt Questions 1–20
- Allow about 30 minutes for this section

Section II – Pages 10–25

80 marks

- Attempt Questions 21–41
- Allow about 1 hour and 30 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:

STUDENT NUMBER/NAME:

Section I

20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

STUDENT NUMBER/NAME:

- 1 At a TAFE college, assessment tasks are graded as: A⁺, A, A⁻, B⁺, etc.

What type of data classification is being used by the TAFE college?

- A. Discrete
- B. Continuous
- C. Nominal
- D. Ordinal

- 2 Which term describes the extent to which a score differs from the mean in a data set?

- A. Range
- B. Standard deviation
- C. Frequency
- D. Cumulative frequency

- 3 How many significant figures has the measurement 0.025×10^4 ?

- A. 2
- B. 3
- C. 4
- D. 6

- 4 Benjamin has a credit card with an outstanding balance of \$485.

Benjamin's bank charges 0.06% compound interest per day on outstanding balances.

How much interest should Benjamin be charged on his outstanding balance over the next 20 days?

- A. \$3.21
- B. \$5.85
- C. \$488.21
- D. \$490.85

- 5 A sheep farmer sells bales of wool in Grade A at \$6.50/bale and Grade B at \$9.50/bale.

If a customer buys more than 100 bales in total, the farmer reduces the total cost of the purchase by 15%.

The owner of a local mill in town buys 45 bales of Grade A and 75 bales of Grade B wool from the farmer.

What is the total price paid by the mill owner?

- A. \$777.50
- B. \$854.25
- C. \$1005
- D. \$1185

- 6 Which of the following costs is NOT included in the sale price of a new motor vehicle?

- A. Stamp duty
- B. Registration
- C. Third party insurance
- D. Comprehensive insurance

- 7 Janice measures the thickness of a piece of foam rubber packing with a ruler calibrated in millimetres.

She recorded her measurement as 4.4 cm.

What is the percentage error of Janice's measurement?

- A. ± 0.114
- B. ± 0.05
- C. ± 1.14
- D. ± 11.4

- 8 City *A* has a Coordinated Universal Time (UTC) of +8 and is currently 4:00 pm.

City *B* is currently 2:00 am on the same day as City *A*.

What is the UTC of City *B*?

- A. -10
 - B. -6
 - C. -2
 - D. +2
- 9 An estate agency charges 3% commission of the first \$250 000 of the sales price of a property, 2% of the next \$400 000 and 1.5% of the remaining sales price.

When a property with a sales price of \$680 000 was sold, what amount of commission does the agency make?

- A. \$12 500
 - B. \$13 250
 - C. \$13 750
 - D. \$15 950
- 10 How many cubic centimetres are contained in a cubic metre?

- A. 10^2
- B. 10^3
- C. 10^6
- D. 10^8

- 11 What is the solution to the equation: $2\sqrt{x+1} = 1$?

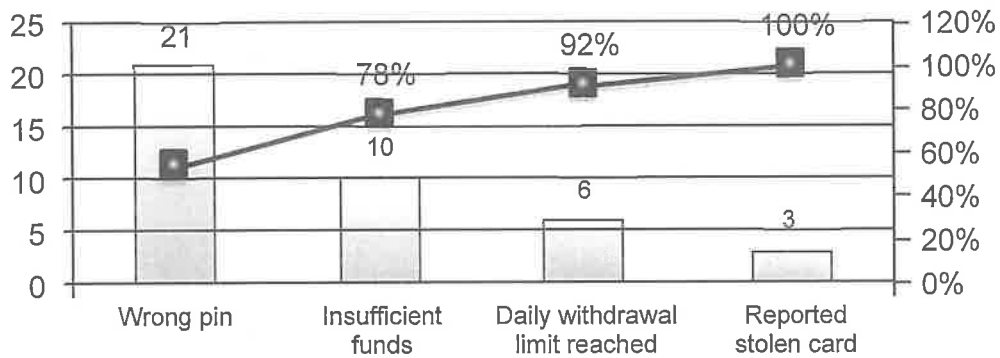
- A. -0.75
- B. -0.25
- C. 0.25
- D. 0.5

- 12 A local Council has increased land rates twice over the last three years. The first was a 5% increase two years ago and the second was a 10% increase last year.

What was the total rate increase (%) over the last three years?

- A. 10.5
B. 15
C. 15.5
D. 50
- 13 One of four reasons are given when the attempted withdrawal of funds from an automated teller machine (ATM) is unsuccessful.

The Pareto chart below gives data from a particular ATM following a number of unsuccessful withdrawal attempts.



What percentage of attempted withdrawals were due to insufficient funds?

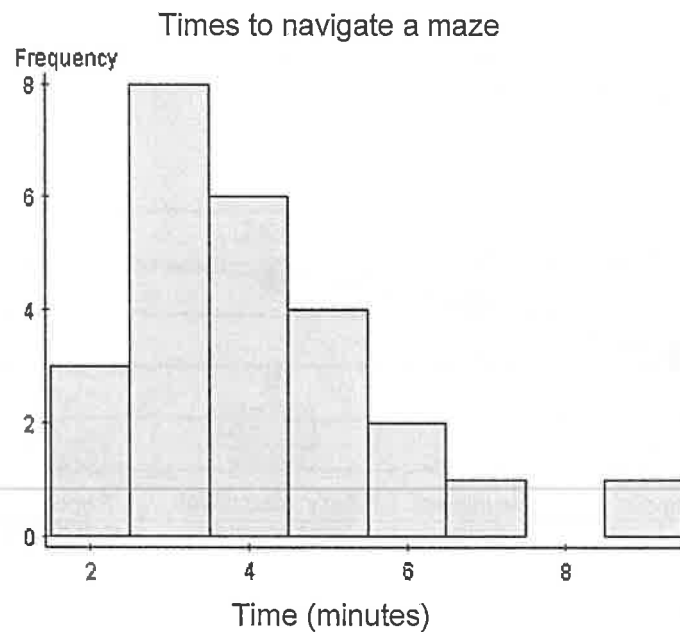
- A. 10
B. 25
C. 46
D. 78

- 14 Two packages are delivered to a residence with an average weight of 1.125 kg.

If one of the packages weighs 750 g, what is the weight of the other package?

- A. 950 g
- B. 1.4125 kg
- C. 1.5 kg
- D. 1.85 kg

- 15 The histogram below shows the times that 25 rats in an animal behaviour study took to successfully navigate a maze.



Which of these statements correctly describes the data distribution?

- A. Bimodal and negatively skewed
- B. Bimodal and positively skewed
- C. Unimodal and negatively skewed
- D. Unimodal and positively skewed

- 16 A male person's Blood Alcohol Content (BAC) can be estimated by the formula:

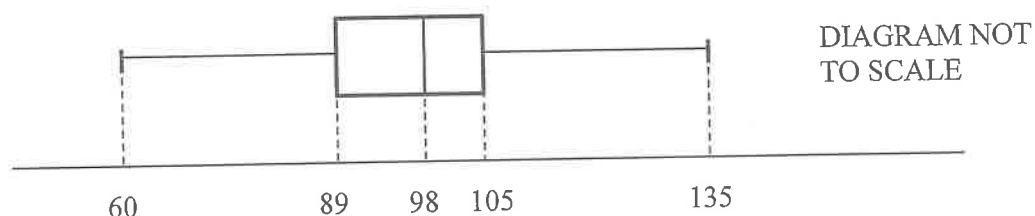
$$BAC = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed in H hours, and M is the person's weight (kg).

Jason consumes two glasses of an alcoholic drink, each containing the equivalent of 1.6 standard drinks, over a period of 45 minutes.

If Jason weighs 70 kg, what would be Jason's approximate BAC (correct to 2 decimal places) at the end of the 45 minutes?

- A. 0.03
B. 0.04
C. 0.05
D. 0.06
- 17 The box and whisker plot below shows the pulse rates (beats/min) of 4 people after exercise at a local fitness centre.

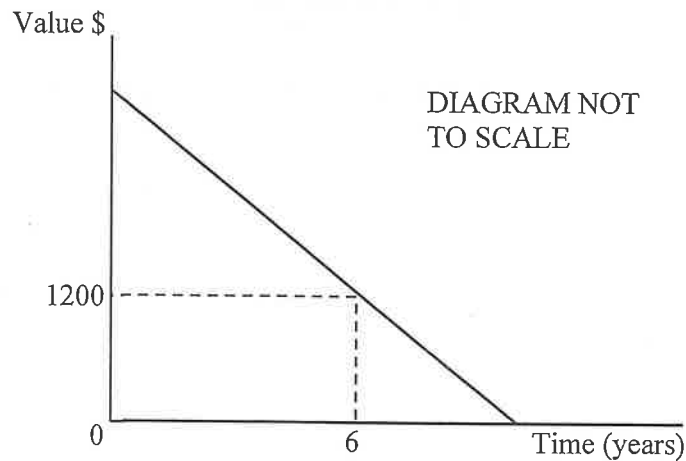


Which of these pulse rates could be considered an outlier for the recorded data?

- A. 64
B. 85
C. 105
D. 128
- 18 What is the perimeter of a semicircular piece of cardboard with diameter 12 cm?

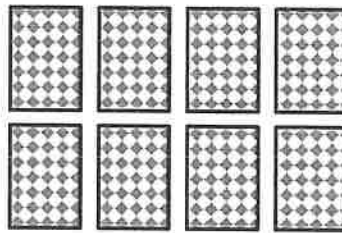
- A. $6(\pi + 2)$
B. $6\pi + 2$
C. $12\pi + 2$
D. $12(\pi + 1)$

- 19 The graph shows that a piece of office equipment has a depreciated value of \$1200 after 6 years.



If the annual rate of depreciation is 12.5%, what was the original purchase price of the office equipment?

- A. \$2400
 B. \$3600
 C. \$3850
 D. \$4800
- 20 Four pairs of cards with either squares, triangles, circles or trapeziums are shuffled and placed on a table with the shape facing down.



When two cards are selected at random and turned over, what is the probability that they form a matching pair of shapes?

- A. $\frac{1}{56}$
 B. $\frac{1}{28}$
 C. $\frac{1}{4}$
 D. $\frac{1}{7}$

Section II**80 marks****Attempt Questions 21–41****Allow about 1 hour and 30 minutes for this section**

Answer the questions in the spaces provided, using a black pen.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the examination. If you use this space, clearly indicate which question you are answering.

Question 21 (1 marks)**Marks**

The mass of the planet Jupiter is approximately 1898×10^{24} kg.

1

This number when written in standard form is 1.898×10^x .

What is the value of x ?

.....

Question 22 (2 marks)

A bucket is being filled with water according to the linear equation $V = 12.5 + 2.5t$,

2

where V is the volume of water in the bucket (litres) and t is the time in minutes.

After the first three minutes of filling, the bucket is half full.

What is the capacity of the bucket?

.....

.....

STUDENT NUMBER/NAME:

Question 23 (2 marks)

Marks

The medicine dosage for a child in comparison with the adult dosage can be determined by the formula:

2

$$D = \frac{MA}{150}$$

where: D is the child dosage (mL)
 M is the age of the child in months
 A is the equivalent adult dosage (mL)

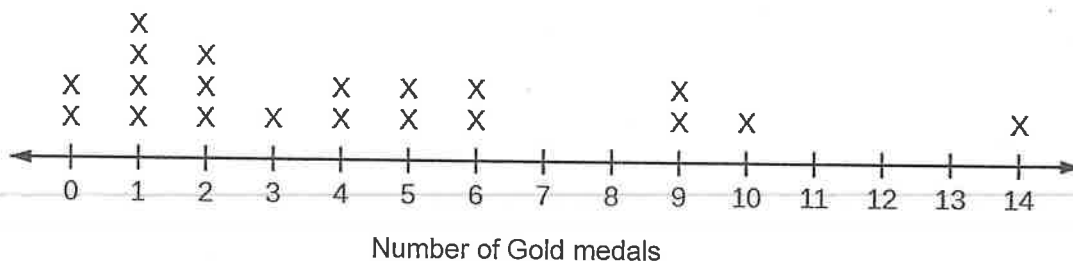
If the equivalent adult dosage is 20 mL, what is the dosage of medicine for a 3-year-old child?

.....

.....

Question 24 (6 marks)

The number of gold medals won by 20 countries in a particular Olympic game, is shown in the following data plot.



- (a) What was the mode number of gold medals won?

1

.....

- (b) What percentage of the countries won more than 4 medals?

2

.....

- (c) Complete a 5 number summary from the data presented.

2

.....

.....

- (d) What is the inter-quartile range?

1

.....

.....

Question 25 (2 marks)**Marks**

Zara is paid at an hourly rate of \$36.50 and worked 25 hours at this rate last week.

2

Zara also worked some extra hours at the time-and-a-half rate last week.

If she was paid a total of \$1131.50 for her work last week, how many extra hours did she work?

.....

.....

.....

Question 26 (2 marks)

A lamp shade in the shape of a hemisphere has a diameter of 30 cm.

2

Calculate the outside surface area of the lamp shade to the nearest whole number.

.....

.....

.....

Question 27 (2 marks)

A formula for finding the sum S of integers is: $S = \frac{n(2a + (n-1)d)}{2}$

2

where: a is the first integer, d is the difference between the integers and n is the number of integers.

Use the formula to calculate the sum of the first 150 even positive integers:

$$2 + 4 + 6 + 8 + \dots + 300$$

.....

.....

.....

STUDENT NUMBER/NAME:

Question 28 (2 marks)

Marks

A person's breakfast has a total of 1445 kilojoules (KJ) which is about 17% of an average person's daily KJ total.

2

If 1 KJ = 0.2 calories, approximately what is the average person's daily calorie intake?

.....

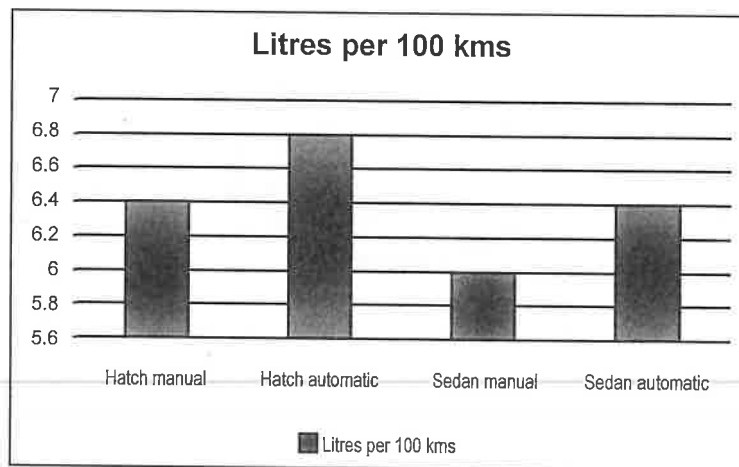
.....

.....

Question 29 (3 marks)

The following chart compares the average fuel consumption of 4 different models of a new vehicle.

3



Janet owns an automatic sedan which has a capacity of 60 litres.

On her last trip using a full tank of petrol, Janet travelled 857 km.

By what percentage did the fuel consumption on her trip differ from the stated average in the chart?

.....

.....

.....

STUDENT NUMBER/NAME:

Question 30 (2 marks)

Marks

Change the subject of the following formula to V :

2

$$r = \sqrt{\frac{3V}{\pi}}$$

.....

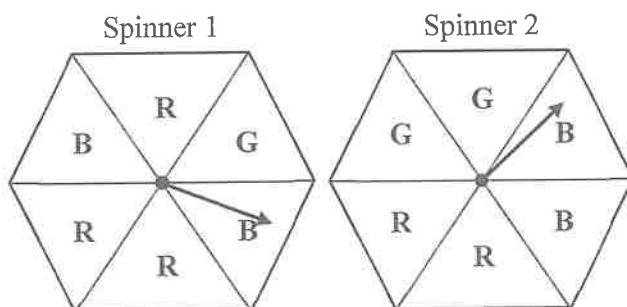
.....

.....

Question 31 (2 marks)

In a probability experiment, Kyle spins one of the following spinners 120 times.

2



The results of the experiment are shown in the table below.

<i>Colour</i>	<i>Frequency</i>
Red (R)	56
Blue (B)	38
Green (G)	26

Based on the results, which spinner did Kyle more likely spin? Give an explanation for your answer.

.....

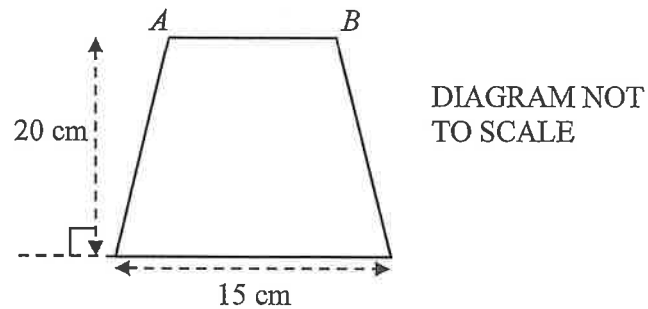
.....

.....

.....

Question 32 (4 marks)**Marks**

A trapezium has dimensions as shown in the following diagram.

2

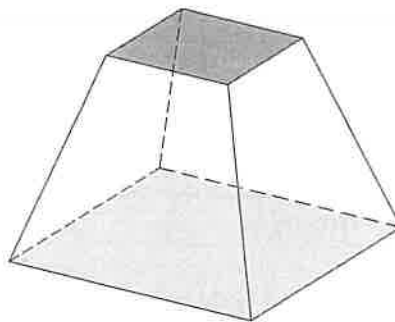
- (a) If the area of the trapezium is 270 cm^2 , show that the length of the top edge AB is 12 cm.

.....

.....

.....

- (b) A three dimensional solid has been made using 4 of these trapeziums and a top and base, as shown in the diagram below.

2

What is the surface area of the solid?

.....

.....

.....

STUDENT NUMBER/NAME:

Question 33 (5 marks)

Marks

A company, *A*, hires cranes for building sites and charges a flat fee of \$650 plus \$125 per hour while the crane is on site.

A second company, *B*, charges a flat fee of \$425 plus \$140 per hour while the crane is on site.

- (a) The cost equation for company *A*'s hiring charges is: $C = 650 + 125h$ where h is the number of hours the crane is on site. 1

Write an equivalent cost equation for company *B*'s hiring charges.

.....

- (b) For how many hours would both companies charge the same total amount for hiring their cranes? 2

.....

.....

.....

- (c) A building site needs to hire a crane for 12 days. 2

From which of the two companies would the total hiring charge be cheaper?

.....

.....

.....

STUDENT NUMBER/NAME:

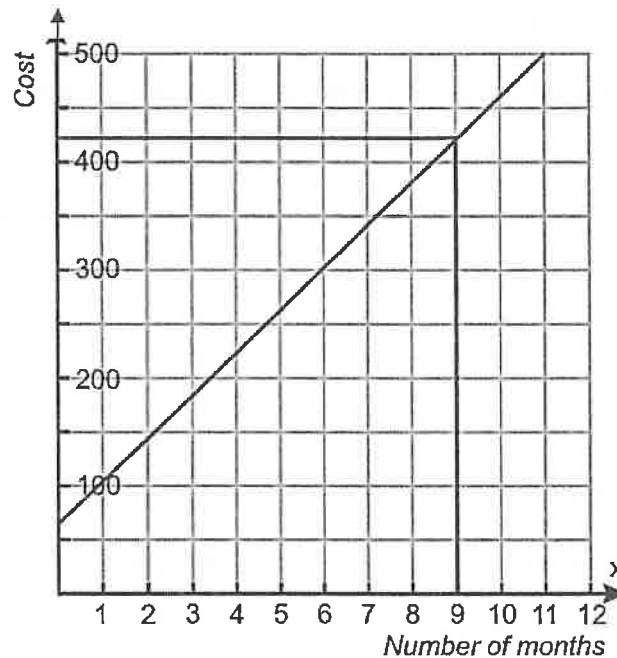
Question 34 (7 marks)

Marks

Marcus has purchased a mobile phone agreeing to pay a \$75 deposit and an equal amount each month until the phone is paid in full.

The graph below shows the cost of the phone over a number of monthly payments

Cost of a mobile phone



- (a) After how many months has Marcus paid a total of \$300 for his phone? 1
.....
- (b) What amount is Marcus paying per month for his phone after his deposit? 1
.....
- (c) Write an equation representing the linear relationship shown in the graph. 2
.....
- (d) How much has Marcus paid for his phone after 9 months? 1
.....
- (e) Marcus pays a total of \$750 for his phone. 2
How many months has he taken to pay the phone off?
.....

Question 35 (6 marks)**Marks**

The following tax table gives the rates of tax payable on categories of taxable income.

<i>Taxable income \$</i>	<i>Tax payable</i>
0 – 18 200	nil
18 201 – 37 000	19 cents for each \$1 over \$18 200
37 001 – 87 000	\$3572 plus 32.5 cents for each \$1 over \$37 000
87 001 – 180 000	\$19 822 plus 37 cents for each \$1 over \$87 000
180 001 and over	\$54 232 plus 45 cents for each \$1 over \$180 000

- (a) Define *taxable income*. 1
-
-
- (b) What is the percentage rate of tax paid on incomes over \$87 000? 1
-
- (c) Determine the tax payable on a taxable income of \$125 000. 2
-
-
- (d) Demetre has a taxable income of \$125 000 and has paid tax by the P.A.Y.E method of \$36 250 during the last financial year. 2

Determine whether Demetre is entitled to a tax refund or has a tax debt and calculate this amount.

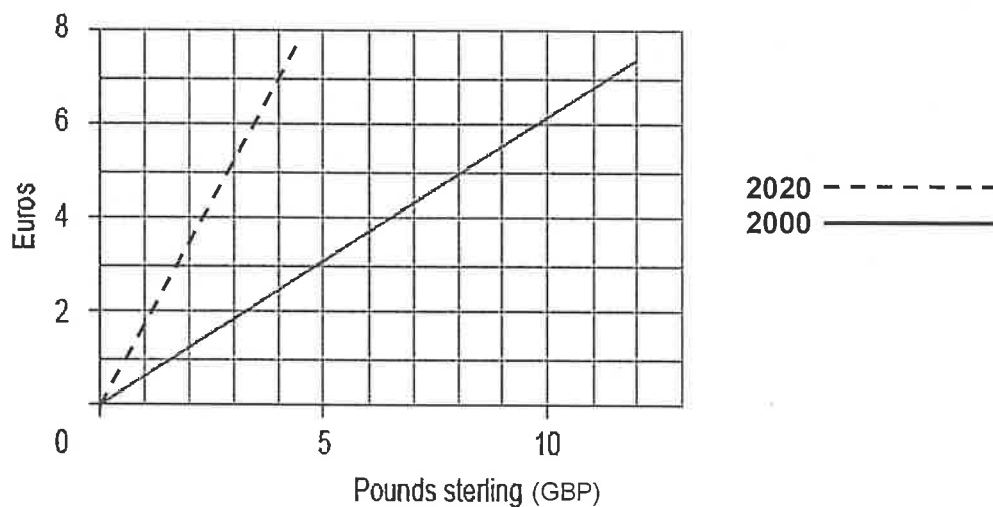
.....

.....

.....

Question 36 (6 marks)**Marks**

The linear conversion graph shows the comparison of the Euro dollar (EUR) with the British Pound Sterling (GBP) in the year 2000 and in 2020.



- (a) Reading from the graph, in the year 2000, 4 GBP could buy 2.5 Euro dollars (Euros). 1

Complete the exchange rate between the two currencies for the year 2000.

1 GBP = EURO

- (b) What was the exchange rate in the year 2020? 2

1 GBP = EURO

- (c) Approximately how many GBP would 1000 EUR have been equivalent to in the year 2020? 1

.....

- (d) In 2020, 1 GBP was equivalent to 1.89 Australian dollars (AUD). 2

What was the exchange rate between AUD and EUR in 2020?

.....

.....

STUDENT NUMBER/NAME:

Question 37 (5 marks)

Marks

The following table shows the percentage of the population within each category of blood type.

<i>Blood type</i>	<i>Percentage of population</i>
O ⁺	40
O ⁻	9
A ⁺	31
A ⁻	7
B ⁺	8
B ⁻	2
AB ⁺	2
AB ⁻	1

Two patients visit their local doctor and have blood tests.

What is the probability that:

- (a) Both patients tested had the most common blood type? 1

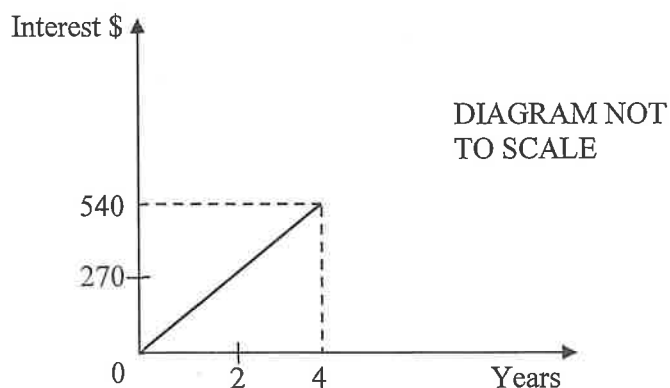
- (b) One of the patients tested had the most common blood type? 2

- (c) At least one of the patients tested had the rarest blood type? 2

Question 38 (7 marks)**Marks**

An amount of \$3000 is invested in an account paying a flat rate of interest.

The graph shows the interest earned on the investment over a period of 4 years.



- (a) Show that the annual rate of interest earned on the investment was 4.5%. 2
-
-
-
- (b) Write the equation representing the graph using I for Interest and n for the number of years. 1
-
- (c) How many years will it take for the investment to attract a total of \$877.50 in interest at the annual rate of 4.5%? 1
-
-
- (d) The initial investment and interest earned over the first 4 years is reinvested for a further 2 years at an annual rate of 3.5%. 3

Calculate the total interest earned over the 6 years?

.....

.....

.....

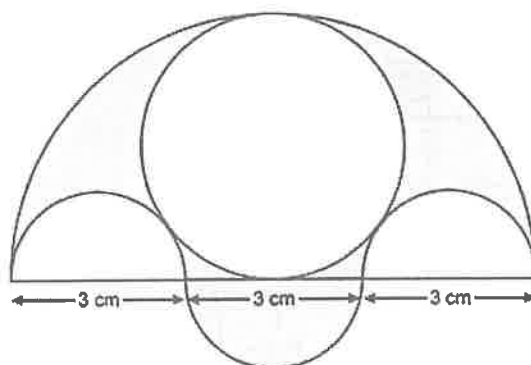
STUDENT NUMBER/NAME:

Question 39 (4 marks)

Marks

In the semicircular design shown below, the large semicircle has diameter 9 cm.

4



What is the total area of the shaded sections of the design? (Give your answer correct to one decimal place).

.....

.....

.....

.....

.....

STUDENT NUMBER/NAME:

Question 40 (4 marks)

Marks

Water is being poured into a large conical tank of height 10 m and radius 4 m, as shown in the diagram.

4

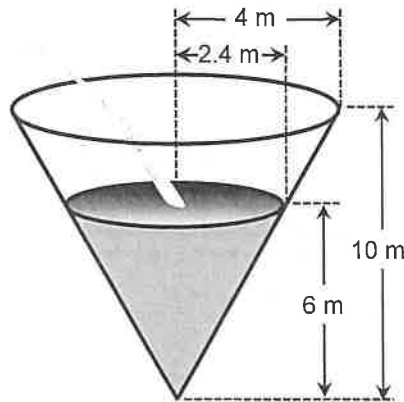


DIAGRAM NOT
TO SCALE

When the water level reaches a height of 6 m in the tank, the radius of the surface of the water is 2.4 m, as shown.

Using the formula $V = \frac{\pi r^2 h}{3}$ calculate the amount of water (to the nearest 10 litres) that still needs to be poured into the tank to reach its capacity.

(Use the conversion 1 cubic metre = 1000 litres in your working).

.....

.....

.....

.....

.....

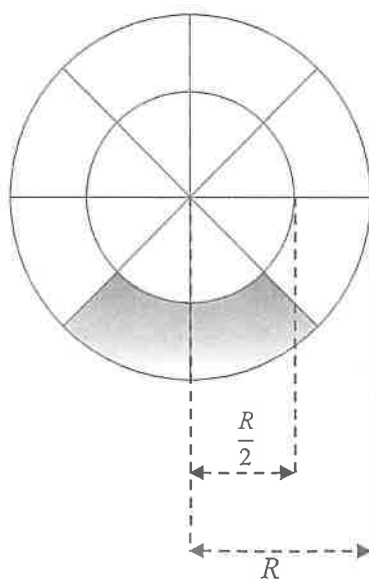
.....

STUDENT NUMBER/NAME:

Question 41 (6 marks)

Marks

Amy has begun drawing a concentric circular design and has shaded a section of it as shown below.



The inner circle has radius $\frac{R}{2}$, the outer circle has radius R , and the circles are divided into 8 equal sectors, as shown.

- (a) What is the centre angle of each of the 8 sectors?

1

.....

- (b) Show that the area (A) of the shaded section of Amy's design can be given by:

3

$$A = \frac{3\pi R^2}{16}$$

.....

.....

.....

- (c) If the shaded section has area 60 cm^2 , calculate the value of R correct to the nearest whole number.

2

.....

.....

.....

End of paper

STUDENT NUMBER/NAME:

Section II – Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of blank white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

Mathematics Standard 1

Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

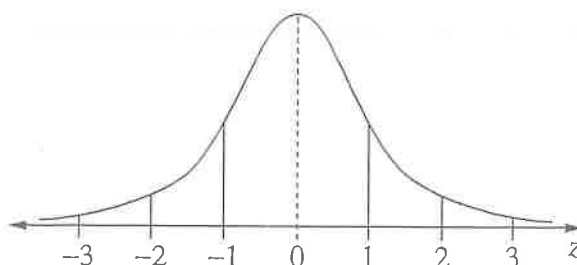
$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

